IMPERIAL

Department of Computing
70015: Mathematics for Machine Learning
Lecturer: Matthew Wicker

Practice Problems 05: Convexity and Convergence

Question 44 (Largest Stepsize for Progress Under L-Smoothness)**.** (III) NB: In an exam context I would give the context we had in the lecture which is necessary to solve this one.

In class we used the smoothness inequality to show, for gradient descent

$$\theta^{(t+1)} = \theta^{(t)} - \alpha \nabla \mathcal{L}(\theta^{(t)}),$$

that with $\alpha = \frac{1}{L}$ we obtain a decrease

$$\mathcal{L}(\theta^{(t+1)}) \leq \mathcal{L}(\theta^{(t)}) - \frac{1}{2L} \left\| \nabla \mathcal{L}(\theta^{(t)}) \right\|^2.$$

What is the *largest* choice of α that still guarantees progress (descent) in one step for any L-smooth loss $\mathcal{L}$?

Question 45 (Tangent convexity)**.** (II) Consider the following definition of convexity which uses the secant of the function:

Definition 5.0.1. A function $f : \mathbb{R} \to \mathbb{R}$ is convex if and only if $\forall x, y \in \mathbb{R}, \forall t \in [0, 1]$ we have that: $f(tx + (1-t)y) \leq tf(x) + (1-t)f(y)$

- Given that the function is everywhere differentiable, state the definition of convexity in terms of the tangent of a function. To do so it can help to draw a convex function first, and then reason about the relationship between the function's values and the tangent line at each point.
- Use your answer to the above to prove that the maximum of two convex function is itself a convex function.

Question 46 (Rayleigh quotient)**.** (III) The *Rayleigh quotient* is defined for a symmetric matrix $\boldsymbol{A} \in \mathbb{R}^{d \times d}$ and a non-zero vector $\boldsymbol{x} \in \mathbb{R}^{d \times 1}$:

$$R(\boldsymbol{A}, \boldsymbol{x}) = \frac{\boldsymbol{x}^\top \boldsymbol{A} \boldsymbol{x}}{||\boldsymbol{x}||_2^2}, \quad ||\boldsymbol{x}||_2^2 = \boldsymbol{x}^\top \boldsymbol{x}.$$

Show that $R(\boldsymbol{A}, \boldsymbol{x}) \in [\lambda_{min}(\boldsymbol{A}), \lambda_{max}(\boldsymbol{A})]$.

This result immediately indicates that $\lambda_{min}(\boldsymbol{A})||\boldsymbol{x}||_2^2 \leq \boldsymbol{x}^\top \boldsymbol{A} \boldsymbol{x} \leq \lambda_{max}(\boldsymbol{A})||\boldsymbol{x}||_2^2$, which is used to prove gradient descent convergence.

Question 47 (Gradient descent with pre-conditioning). (III)

Consider the following update rule named *pre-conditioned gradient descent*:

$$\theta_{t+1} = \theta_t - \gamma_t \boldsymbol{P}_t^{-1} \nabla_\theta L(\theta_t).$$

Here $\boldsymbol{P}_t$ is called *pre-conditioner* at time step t. We consider linear regression as an example, and assume constant learning rate and pre-conditioner, i.e., $\gamma_t = \gamma$ and $\boldsymbol{P}_t = \boldsymbol{P}$ for all t. Show that with an appropriate choice of the pre-conditioner $\boldsymbol{P}$, we can achieve a robust selection of the learning rate γ, i.e., if the selected γ works for an initialisation θ_1, it will also work for all other initialisations.

Hints: you can follow the below steps to solve the question:

1. Work out the pre-conditioned gradient descent update in linear regression, and derive θ_t as a function of θ_1, γ, $\boldsymbol{P}$ and the dataset $(\boldsymbol{X}, \boldsymbol{y})$;

2. For a given $\boldsymbol{P}$, work out the learning rates γ_{min} and γ_{max} such that pre-conditioned gradient descent converges when $\gamma < \gamma_{min}$, or diverges when $\gamma \geq \gamma_{max}$;

3. Select $\boldsymbol{P}$ such that $\gamma_{min} = \gamma_{max}$, therefore there exist no interval (like $[\gamma_{min}, \gamma_{max})$) such that convergence depends on initialisation when γ falls into such interval.

Question 48 (Momentum gradient descent). (III) Consider the following update rule named *momemtum gradient descent*, with constant learning rate γ and momentum step-size α:

$$\theta_{t+1} = \theta_t - \gamma \nabla_\theta L(\theta_t) + \alpha \Delta\theta_t,$$

$$\Delta\theta_{t+1} = \theta_{t+1} - \theta_t, \quad \Delta\theta_0 = \mathbf{0}.$$

Show that solving linear regression using momemtum gradient descent, if converges, converges to $\theta^* = (\boldsymbol{X}^\top \boldsymbol{X})^{-1} \boldsymbol{X}^\top \boldsymbol{y}$.

Hint: follow the below steps and practice your linear algebra skills :)

1. Write down the update equations for the parameters θ_t and the momentum $\Delta\theta_t$;

2. Collect both terms as a long vector $(\theta_t^\top, \Delta\theta_t^\top)^\top$, and merge the two linear update equations in step 1 into one "joint" linear equation using block matrices;

3. Apply the analysis techniques for gradient descent convergence for linear regression to show the converged solution (if converges).